SHEEHAN'S CONJECTURE FOR FOUR-REGULAR SIMPLE GRAPHS OF TREEWIDTH AT MOST FIVE

ARGUS OPEN RESEARCH PROJECT

Abstract. Sheehan conjectured in 1975 that no finite simple r-regular graph ($r \geq 3$) possesses a unique Hamiltonian cycle, with 4-regular simple graphs constituting the primary open case. We prove unconditionally that every finite simple 4-regular Hamiltonian graph G of treewidth $\mathrm{tw}(G) \leq 5$ has at least two distinct undirected Hamiltonian cycles. Building on our pointed routing-signature exclusion for $\mathrm{tw}(G) \leq 4$ (bags of size ≤ 5), we establish normal-form join reductions and Hamilton-liveness completion profiles on boundaries of size at most six, and construct an exact operation-closed carrier of 152,441 active pointed states (809,003 construction rows, 88,107,797 positive routing entries, and 25,209,843,635 relative boundary orders accounted with zero uncovered unary or binary targets and zero closed unit markers).

1. Main Mathematical Results

Theorem 1.1 (Exclusion of unique Hamiltonicity for treewidth at most five)**.** *Every finite simple* 4*-regular Hamiltonian graph* G *with* $\mathrm{tw}(G) \leq 5$ *possesses at least two distinct undirected Hamiltonian cycles.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 3167 in the domain of structural graph theory adheres to a strict separation between the exact statement proved and adjacent formulations with stronger or weaker hypotheses:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, number-field, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, finite-field, or symbolic arithmetic without floating-point approximation, and independently re-verified from standalone certificates.

(c) *Resolved scope:* Every finite simple 4-regular Hamiltonian graph of treewidth at most five (and at most four) has at least two distinct undirected Hamiltonian cycles.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that remain open beyond the scope settled above:

- Unrestricted Sheehan's conjecture for 4-regular simple graphs of treewidth >= 6

References

[1] Primary problem list, literature attributions, and exact verification certificates for Result 3167 (structural graph theory), archived in the accompanying repository package `results/3167/`.

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